

closure_evidence_20260922

BOB-198 closure evidence — 2026-09-22

Fix (implements the recorded operator decision exactly — no auth middleware added)

Operator decision (2026-08-26, §11.4.66): “REFUSE TO START LAN-BOUND” — a boot-time guard, not auth middleware.

qBitTorrent-go/cmd/qbittorrent-proxy/main.go: - New checkLoopbackBind(bindHost string) error (line ~163) — refuses anything that is not a loopback address (127.0.0.0/8 literal, ::1, or localhost case-insensitive); syntax-only, no DNS resolution (an unresolvable hostname fails closed rather than being trusted). - Called immediately after log setup in main() (line ~36), before the qBittorrent connection attempt, proxy config, or any store opens. log.Fatal() on refusal — logs the actual configured address and exits non-zero; the code never reaches r.Run(addr). - The listener bind itself changed from the prior bare ":%d" (all-interfaces) form to "%s:%d" using the new ServerBindHost (line ~140) — the real fix, with the boot-time check as defense-in-depth against a future regression.

qBitTorrent-go/internal/config/config.go: new Config.ServerBindHost field, env SERVER_BIND_HOST, defaulting to "127.0.0.1".

No auth middleware was added anywhere — confirmed by inspection, middleware registration (CORS/Logger/GinRateLimit) is unchanged.

RED evidence

The test file was written first, referencing `checkLoopbackBind` before it existed — build failure (undefined: `checkLoopbackBind`) proved the prefix code had no boot-time guard of any kind.

GREEN + golden-FALSE evidence (independently re-verified this session)

```
$ cd qBitTorrent-go && go build ./cmd/qbittorrent-proxy/... ./internal/config/...
$ go vet ./cmd/qbittorrent-proxy/... ./internal/config/...
$ go test ./cmd/qbittorrent-proxy/... ./internal/config/... -v
--- PASS: TestGuard_RefusesNonLoopbackBind (0.00s)
    --- PASS: .../explicit_all-interfaces_literal (0.00s)
    --- PASS: .../empty_host_(Go's_:port_all-interfaces_shorthand) (0.00s)
    --- PASS: .../specific_LAN_IP (0.00s)
    --- PASS: .../another_specific_non-loopback_IP (0.00s)
    --- PASS: .../unresolvable_/_arbitrary_hostname_(fail-closed,_no_DNS_trust) (0.00s)
    --- PASS: .../IPv6_non-loopback_literal (0.00s)
--- PASS: TestGuard_AllowsGenuineLoopbackBind (0.00s)
    --- PASS: .../IPv4_loopback_literal (0.00s)
    --- PASS: .../localhost,_lowercase (0.00s)
    --- PASS: .../localhost,_mixed_case_(case-insensitive) (0.00s)
    --- PASS: .../IPv6_loopback_literal (0.00s)
    --- PASS: .../any_127.0.0.0/8_literal,_not_just_.1 (0.00s)
PASS
ok      github.com/milos85vasic/qBitTorrent-go/cmd/qbittorrent-proxy (cached)
--- PASS: TestLoad_Defaults, TestLoad_EnvOverride, TestLoad_TrackerAuth,
        TestLoad_TrackerCookies, TestLoad_KinozalFallback, TestLoad_MetadataAPIKeys, TestQBittorrentURL (all pre-existing,
        all still pass – no regression from the new ServerBindHost field)
PASS
ok      github.com/milos85vasic/qBitTorrent-go/internal/config (cached)
```

6/6 refusal subtests (0.0.0.0, empty-string all-interfaces shorthand, two specific non-loopback IPs, an arbitrary hostname proving no-DNS-trust, a non-loopback IPv6 literal) and 5/5 golden-FALSE subtests (127.0.0.1, localhost, LocalHost case-insensitivity, ::1, and 127.0.0.53 proving the whole 127.0.0.0/8 block is accepted not just .1) all PASS.

Additional real-process runtime evidence (captured by the implementing agent, beyond what was required)

Built the actual binary and ran it twice on scratch ports: -
SERVER_BIND_HOST=0.0.0.0 → logged the fatal refusal and exited 1;
server never started. - Default (unset) → ss -tlnp confirmed LISTEN
127.0.0.1:<port> (not 0.0.0.0).

git diff –stat

```
qBittorrent-go/cmd/qbittorrent-proxy/main.go | 92 ++++++
+++++
qBittorrent-go/internal/config/config.go      | 20 +++++
2 files changed, 106 insertions(+), 6 deletions(-)
```

Plus new untracked qBittorrent-go/cmd/qbittorrent-proxy/
main_test.go (117 lines).

Residual note (informational, not acted on — out of this item’s scope)

config/lan_route_auth_policy.yaml carries a comment citing this file’s
OLD bind behavior — now stale, but it documents a *different* gate
(scripts/pre_build/check_cm_lan_routes_authenticated.sh, route-level
auth-marker wiring for the Python service) unrelated to this fix; not
touched. docker-compose.yml’s MERGE_SERVICE_HOST=0.0.0.0 for the
qbittorrent-proxy-go service is a distinct env var from the new
SERVER_BIND_HOST this fix introduces and has no effect on this guard.